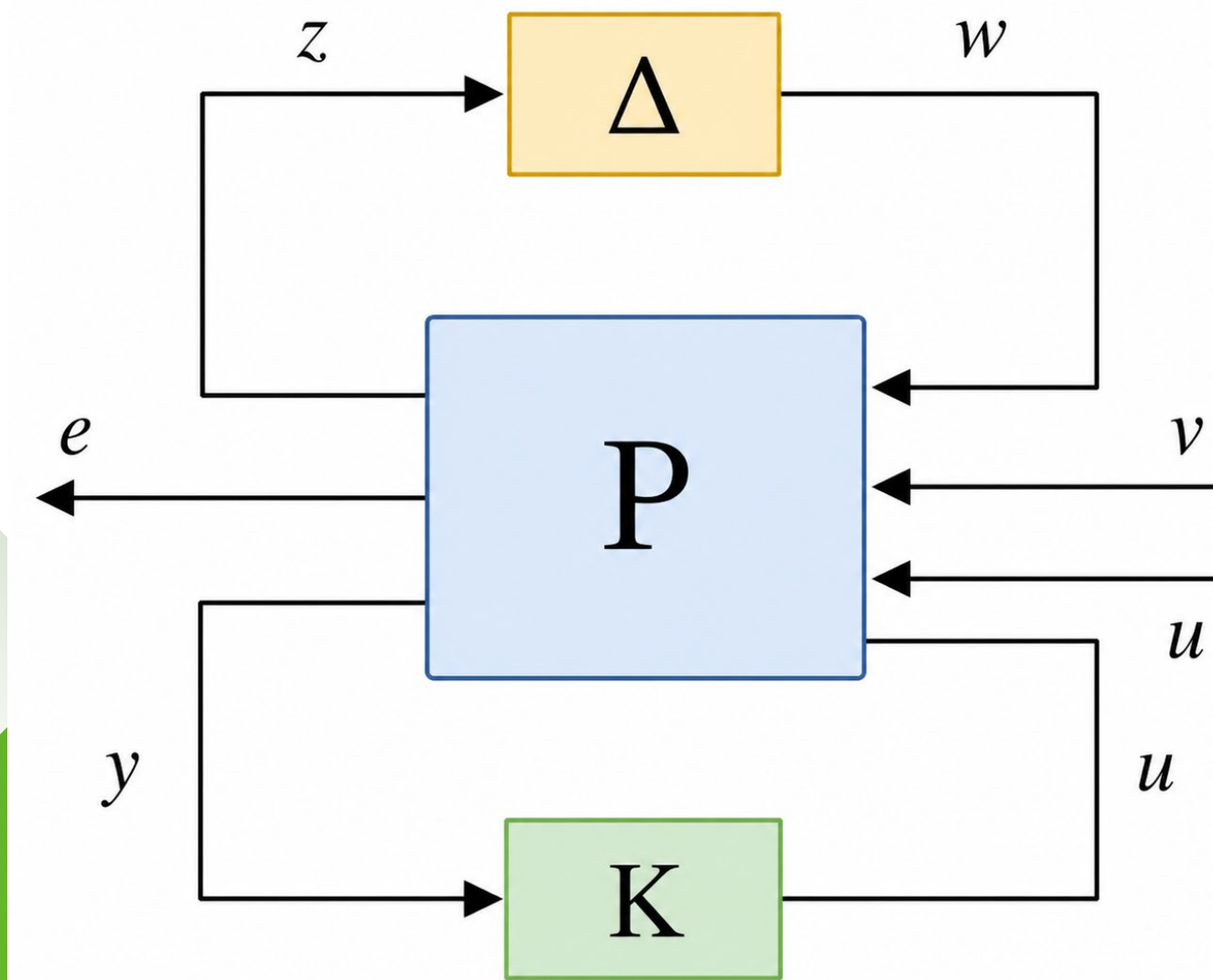


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Introduction to Robust Control (Part 1)



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Preliminares

What is a Robust system ?

What is Robust control?



Introduction

Basic Definitions (informal)

Robust system

A system is "robust" if it is capable of meeting *requirements* in the presence of disturbance or uncertainty.

Robust control

An approach to designing a system that can handle uncertainty.



Introduction

Basic Definitions (informal)





Introduction

Basic Definitions (informal)

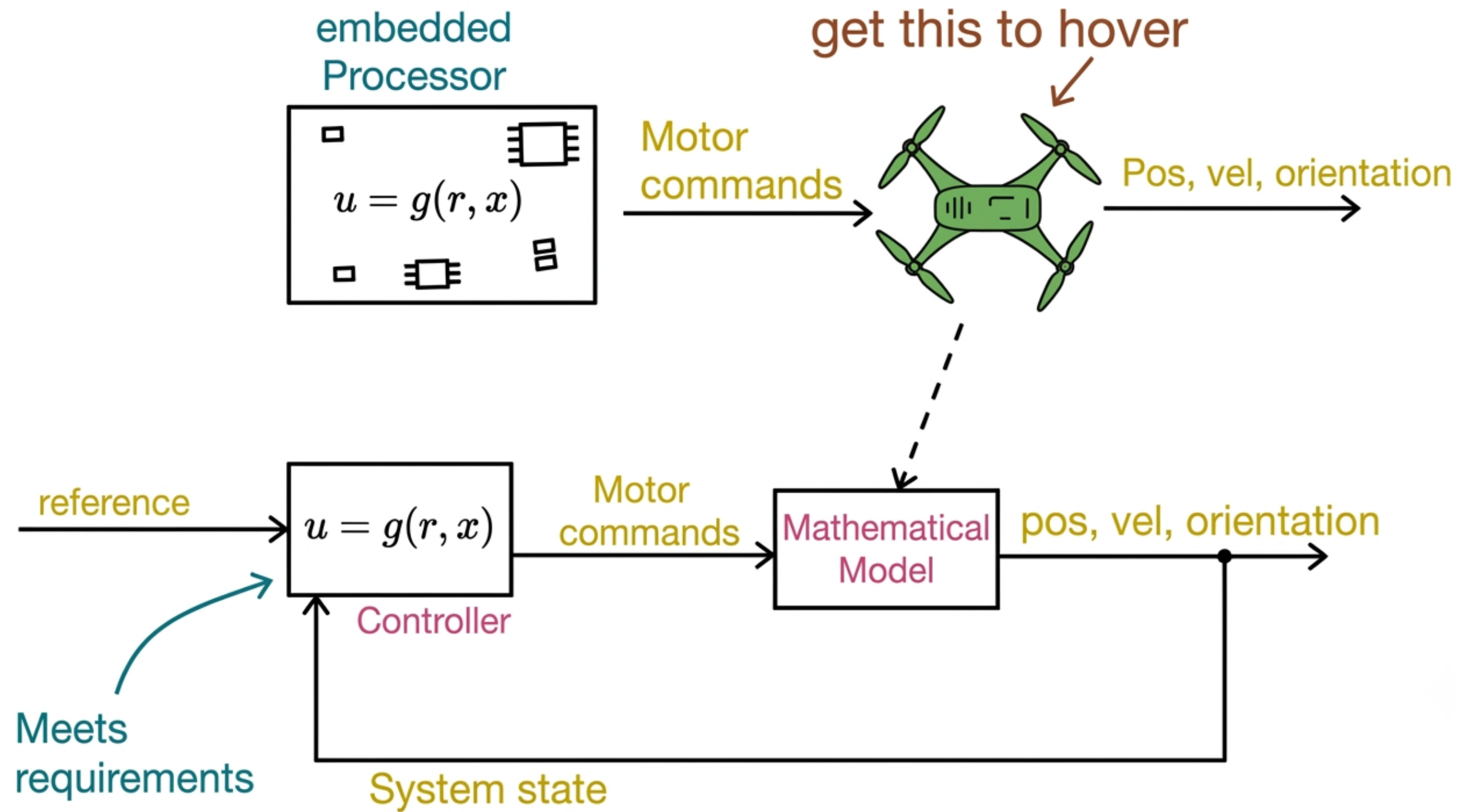
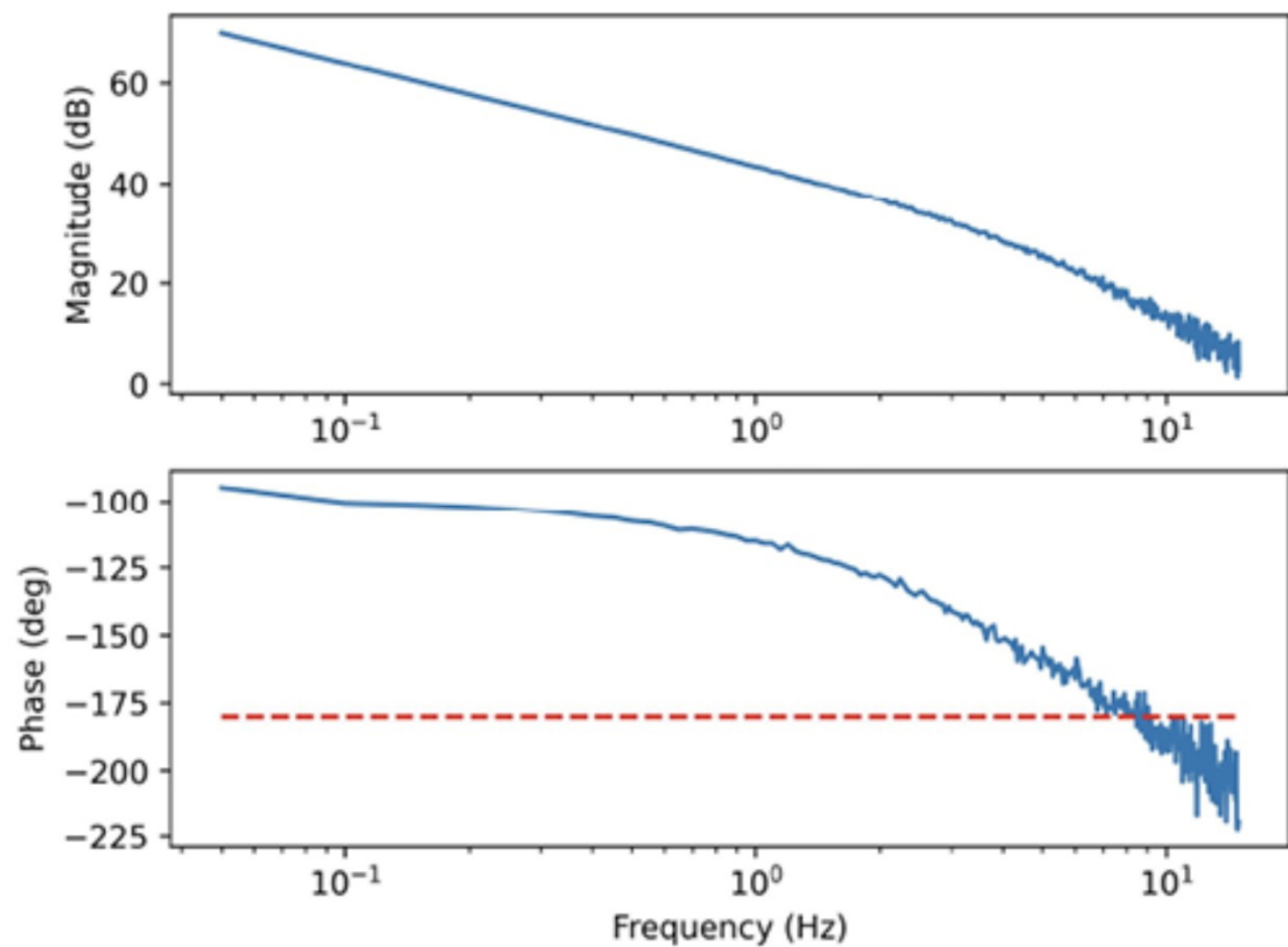


Figure 1: Drone control system methodology.

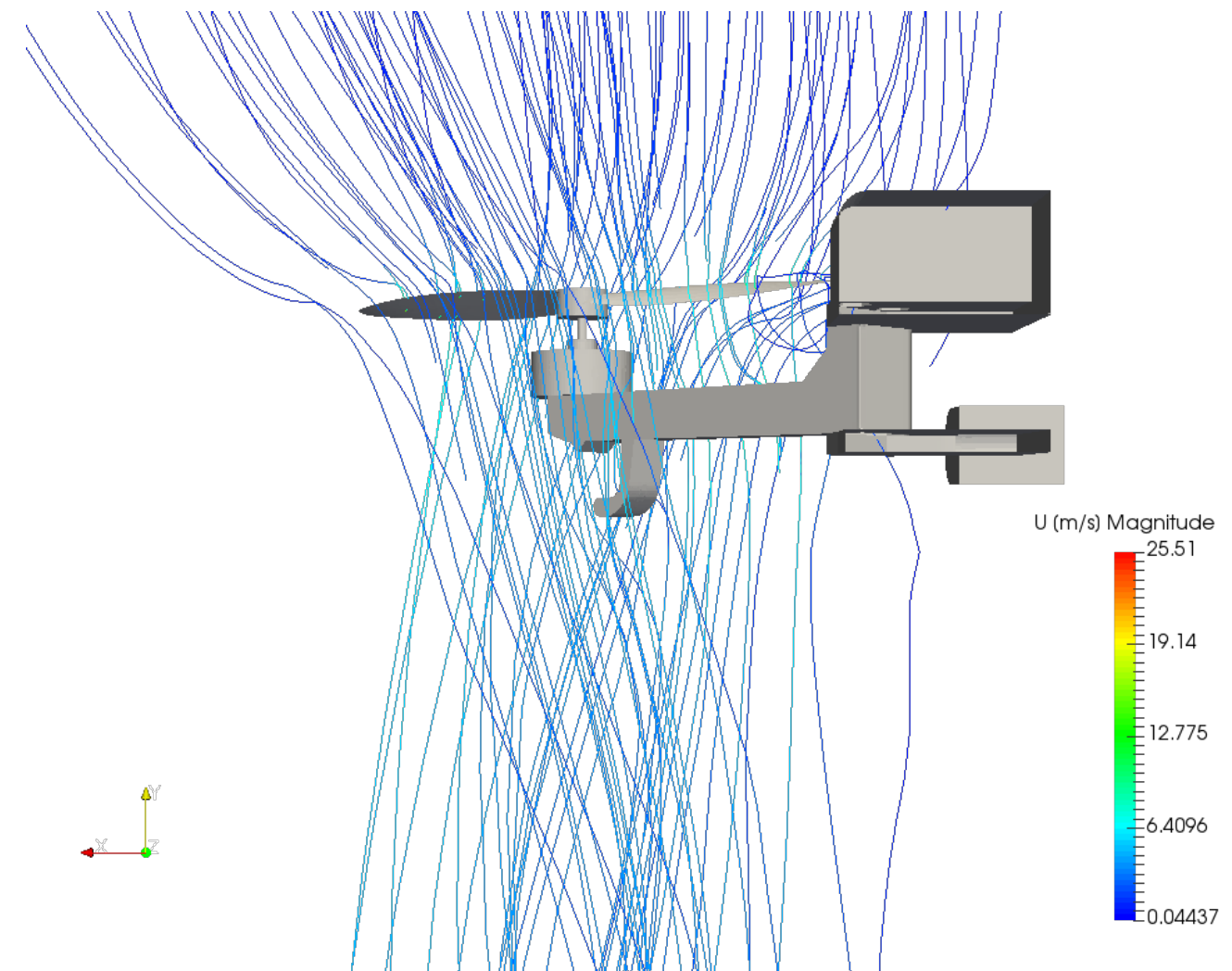


Introduction

Basic Definitions (informal)



(a) High frequency dynamics.



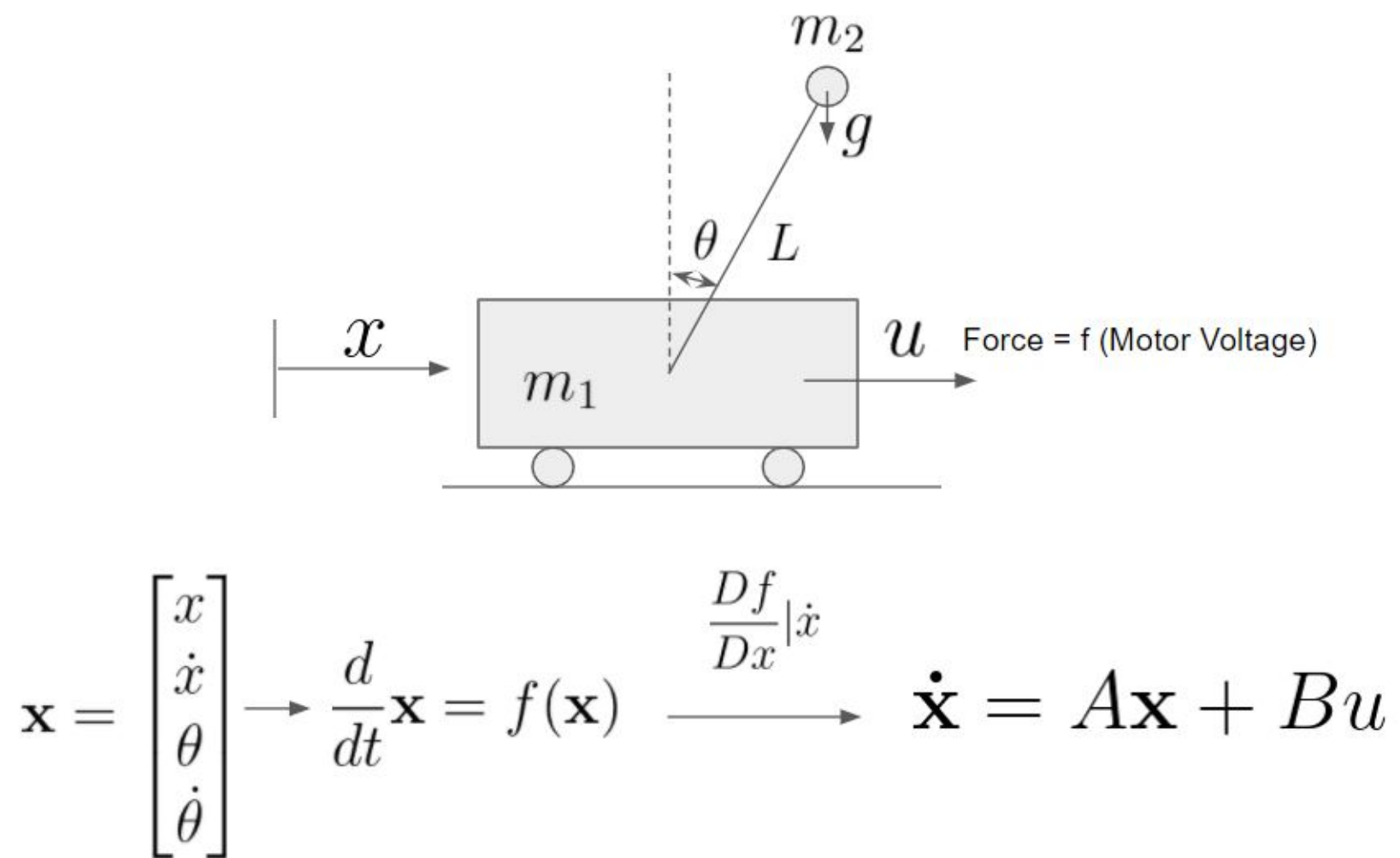
(b) Uncertain driving forces.

Figure 2: Physical systems limitations.

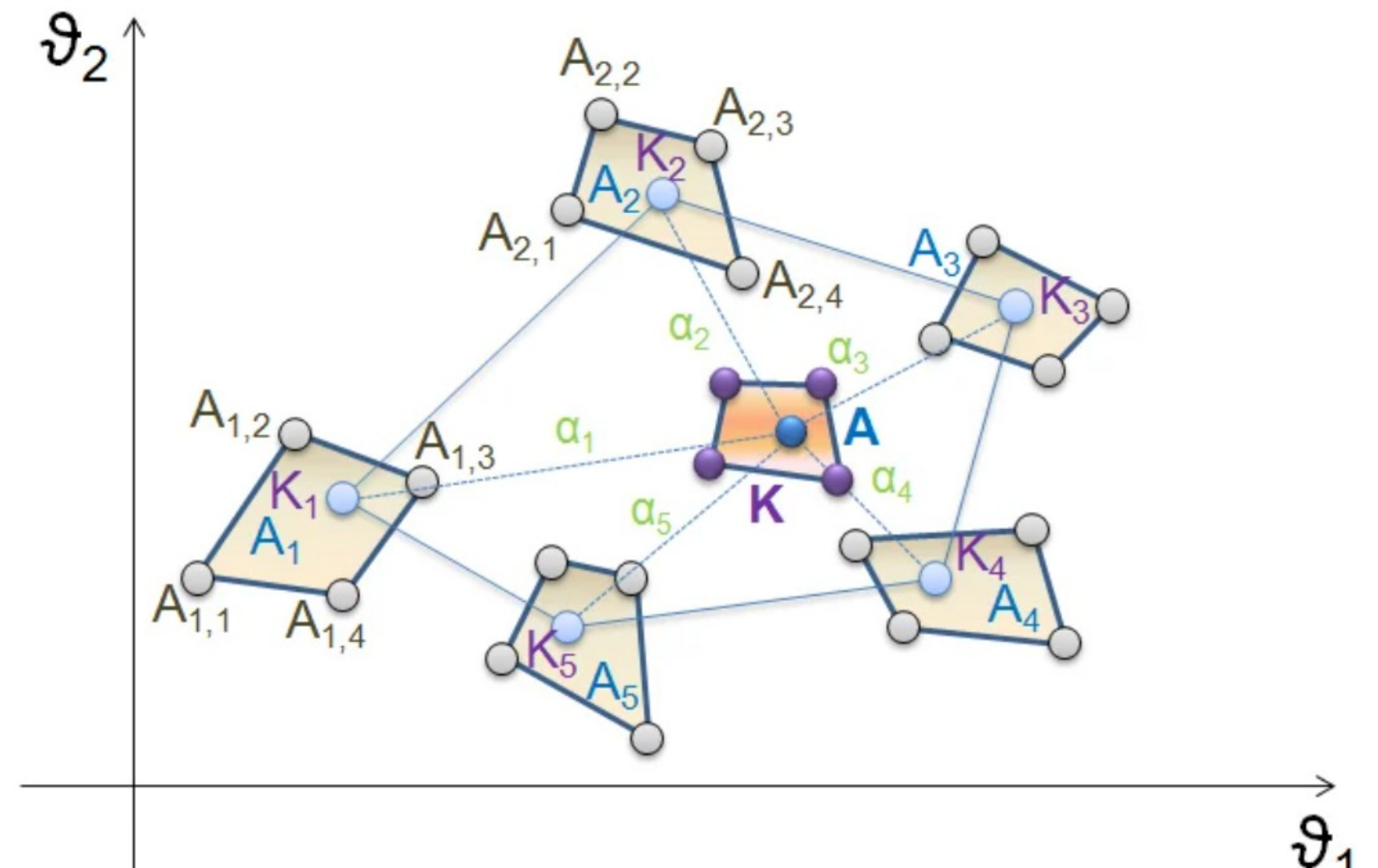


Introduction

Basic Definitions (informal)



(a) Linear models for simplicity.



(b) Natural variations over time.

Figure 3: Physical systems limitations.



Introduction

Basic Definitions (informal)



Figure 4: Manufacturing processes.



$\mathcal{H}_2$ and $\mathcal{H}_\infty$

- The main goal of a control system is to meet **performance specifications** in addition to ensuring **internal stability**. One way to express performance is by measuring the size of certain system signals, such as the tracking error.
- To do this, the Hardy spaces $\mathcal{H}_2$ and $\mathcal{H}_\infty$ are introduced, which provide different norms for measuring signals depending on the context.



$\mathcal{H}_2$ and $\mathcal{H}_\infty$

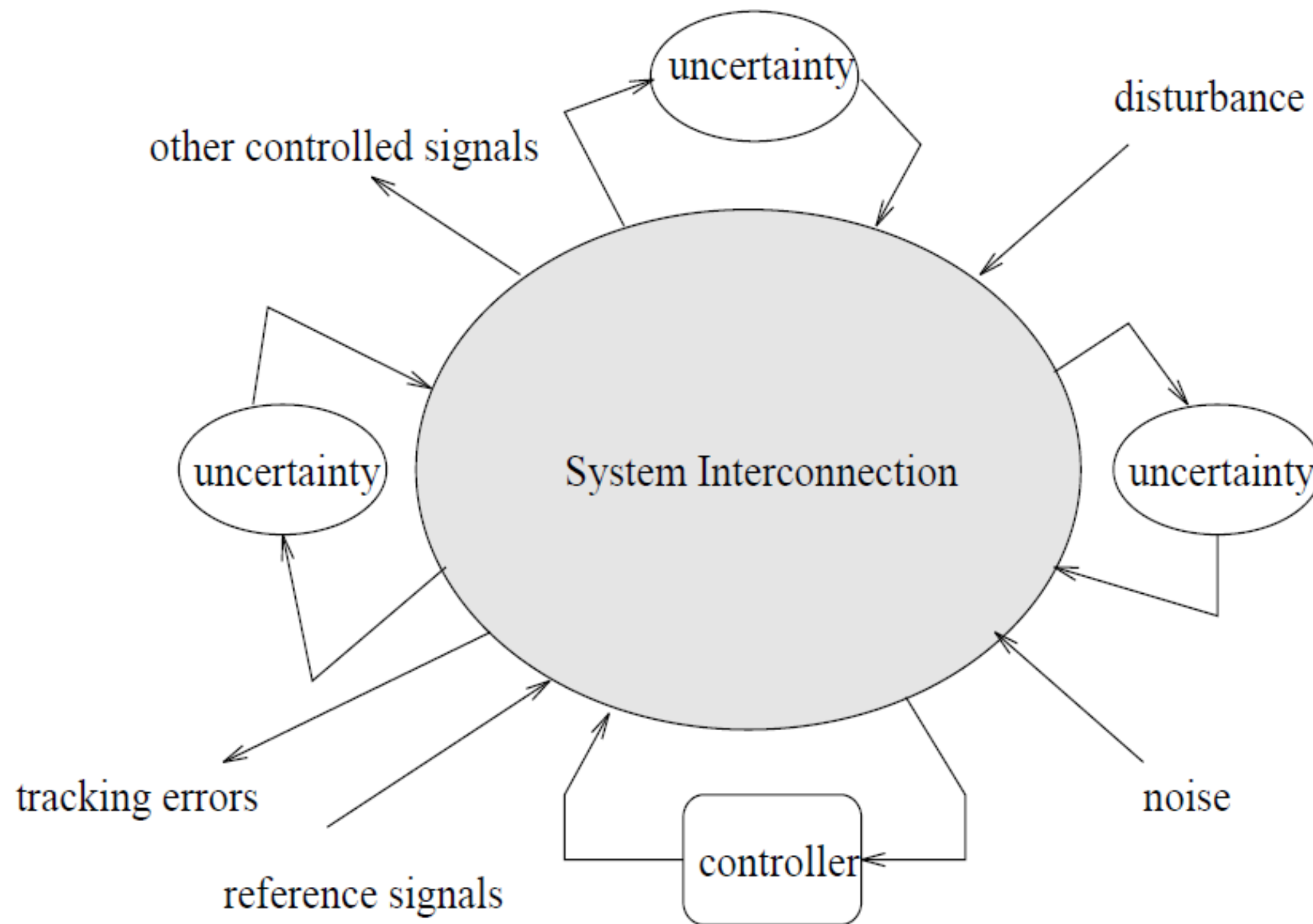


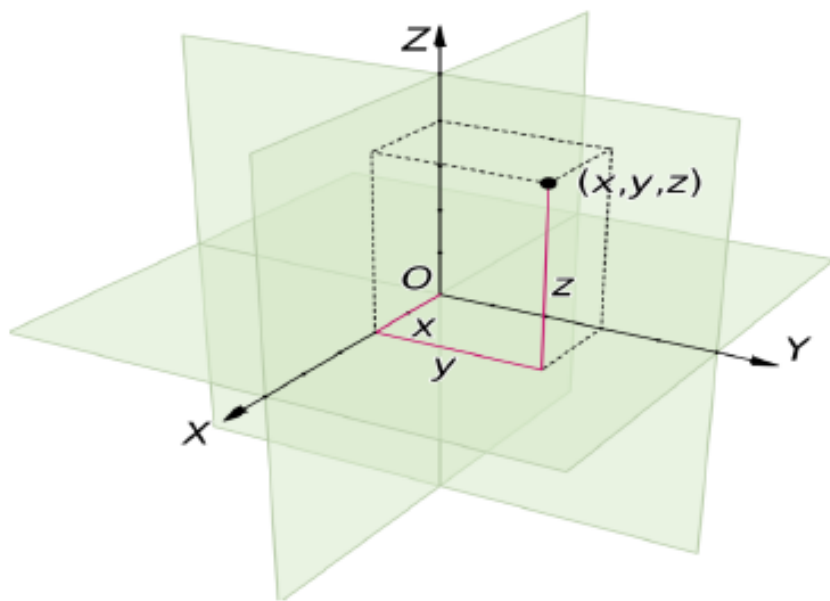
Figure 5: General system interconnection.



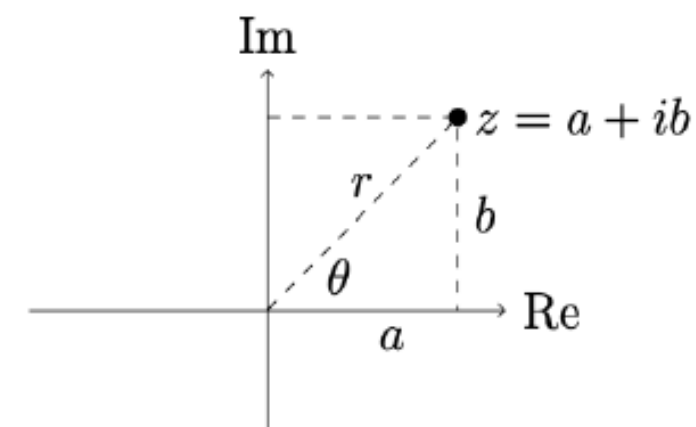
Hilbert Spaces

- Recall the inner product of vectors defined on a Euclidean space $\mathbb{C}^n$:

$$\langle x, y \rangle := x^* y = \sum_{i=1}^n \bar{x}_i y_i \quad \forall x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{C}^n.$$



(a) Euclidean space overview.



(b) Complex vector example.



Hilbert Spaces

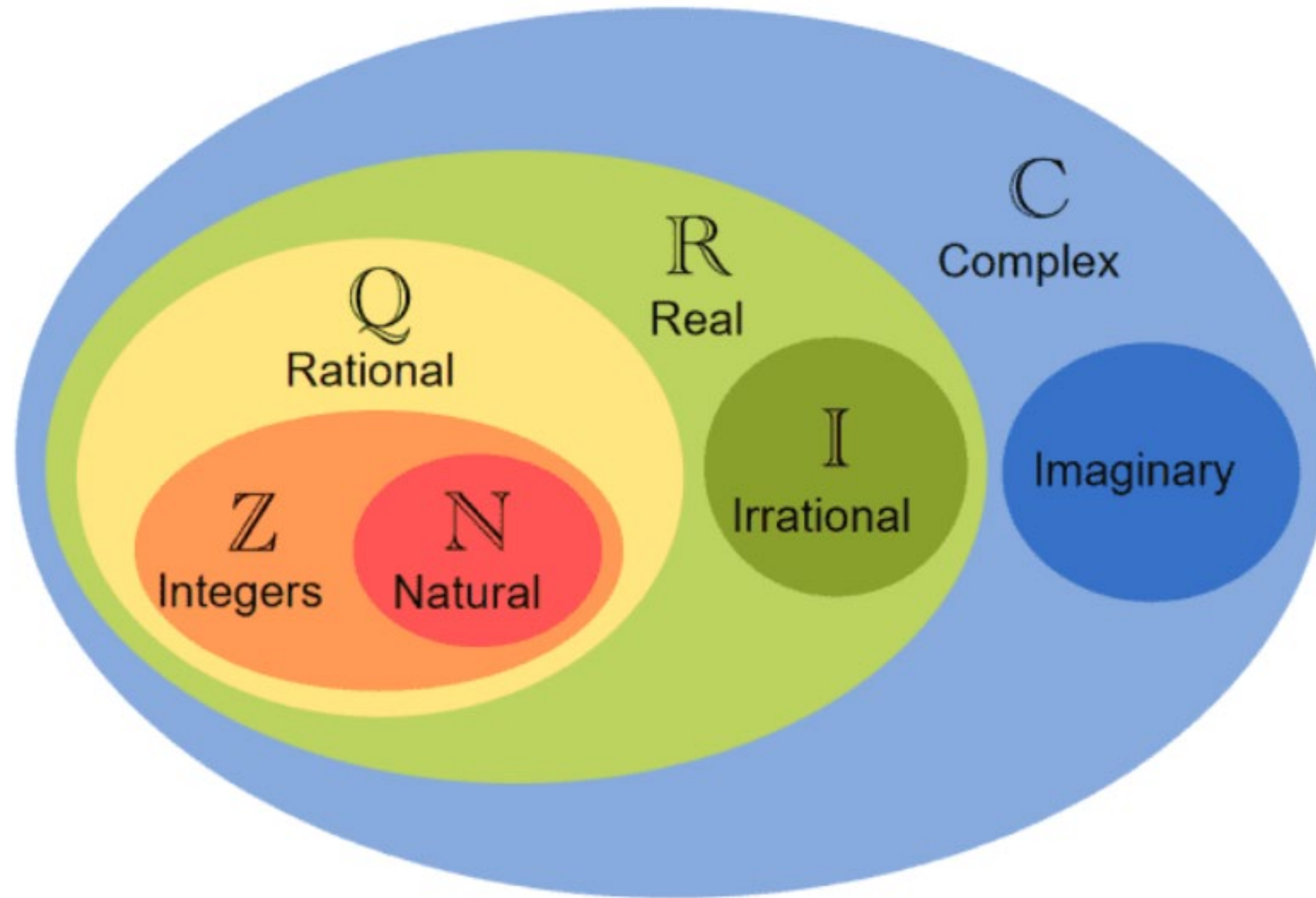


Figure 7: Complex Plane.



Hilbert Spaces

Example 1: Consider the complex vectors x and y and compute the inner product $\langle x, y \rangle$.

$$x = \begin{bmatrix} 1 + i \\ 2 \end{bmatrix}, \quad y = \begin{bmatrix} 3 \\ i \end{bmatrix}$$



Hilbert Spaces

Step 1: Take the conjugate transpose of x

$$x^* = \begin{bmatrix} \overline{1+i} & \overline{2} \end{bmatrix} = \begin{bmatrix} 1-i & 2 \end{bmatrix}$$

Step 2: Compute the inner product $\langle x, y \rangle = x^* y$

$$\langle x, y \rangle = (1-i)(3) + (2)(i) = 3 - 3i + 2i = 3 - i$$

Answer:

$$\boxed{\langle x, y \rangle = 3 - i}$$



Hilbert Spaces

- Many important metric notions and geometrical properties, such as **length, distance, angle**, and the **energy of physical systems**, can be deduced from this inner product. For instance, the length of a vector $x \in \mathbb{C}^n$ is defined as

$$\|x\| := \sqrt{\langle x, x \rangle}$$

- The angle between two vectors

$$\cos(\angle(x, y)) = \frac{\langle x, y \rangle}{\|x\| \|y\|}, \quad \angle(x, y) \in [0, \pi].$$

- Two vectors are said to be orthogonal if $\angle(x, y) = \frac{\pi}{2}$.



Hilbert Spaces

Definition 1. Let V be a vector space over $\mathbb{C}$. An *inner product* on V is a complex-valued function,

$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{C}$$

such that for any $x, y, z \in V$ and $\alpha, \beta \in \mathbb{C}$, the following properties hold:

1. $\langle x, \alpha y + \beta z \rangle = \alpha \langle x, y \rangle + \beta \langle x, z \rangle$
2. $\langle x, y \rangle = \overline{\langle y, x \rangle}$
3. $\langle x, x \rangle > 0$ if $x \neq 0$

A vector space V with an inner product is called an *inner product space*.



Hilbert Spaces

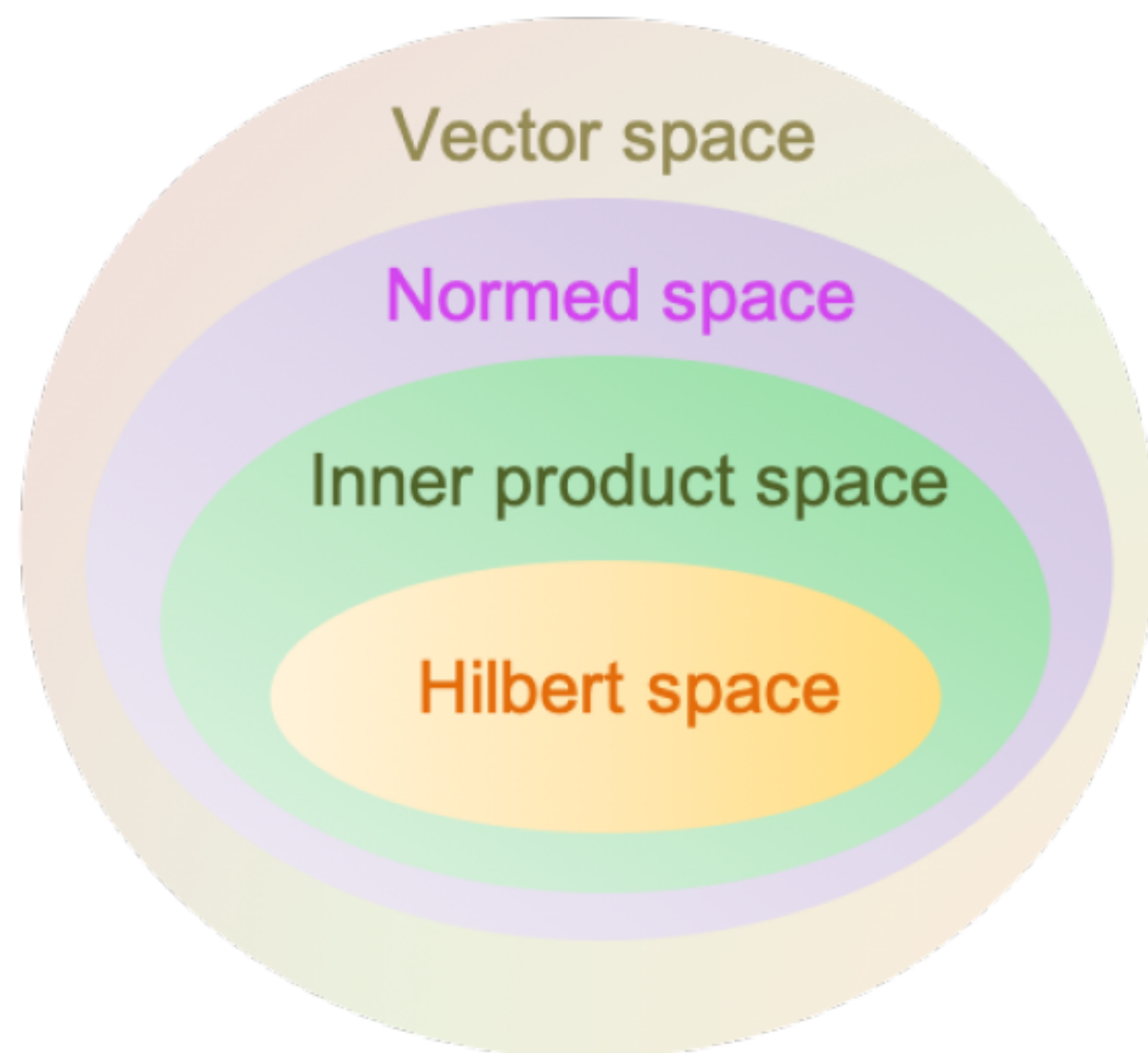


Figure 8: Space vectors diagram

- It is clear that the inner product induces a norm

$$\|x\| := \sqrt{\langle x, x \rangle},$$

- Also, the distance between vectors x and y is

$$d(x, y) = \|x - y\|.$$

- Two vectors x and y in an inner product space V are said to be *orthogonal* if $\langle x, y \rangle = 0$, denoted $x \perp y$.



Hilbert Spaces

- More generally, a vector x is said to be orthogonal to a set $S \subset V$, denoted by $x \perp S$, if $x \perp y$ for all $y \in S$.

The **inner product** and the **inner product induced norm** have the following familiar properties.

Theorem 1. Let V be an inner product space and let $x, y \in V$. Then:

1. $|\langle x, y \rangle| \leq \|x\| \|y\|$ (Cauchy-Schwarz inequality).
2. $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$ (Parallelogram law).
3. $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ if $x \perp y$.



Hilbert Spaces

A **Hilbert space** is a *complete inner product space* with the norm induced by its inner product. For example, $\mathbb{C}^n$ with the usual inner product is a finite dimensional Hilbert space (Recall Example 1).

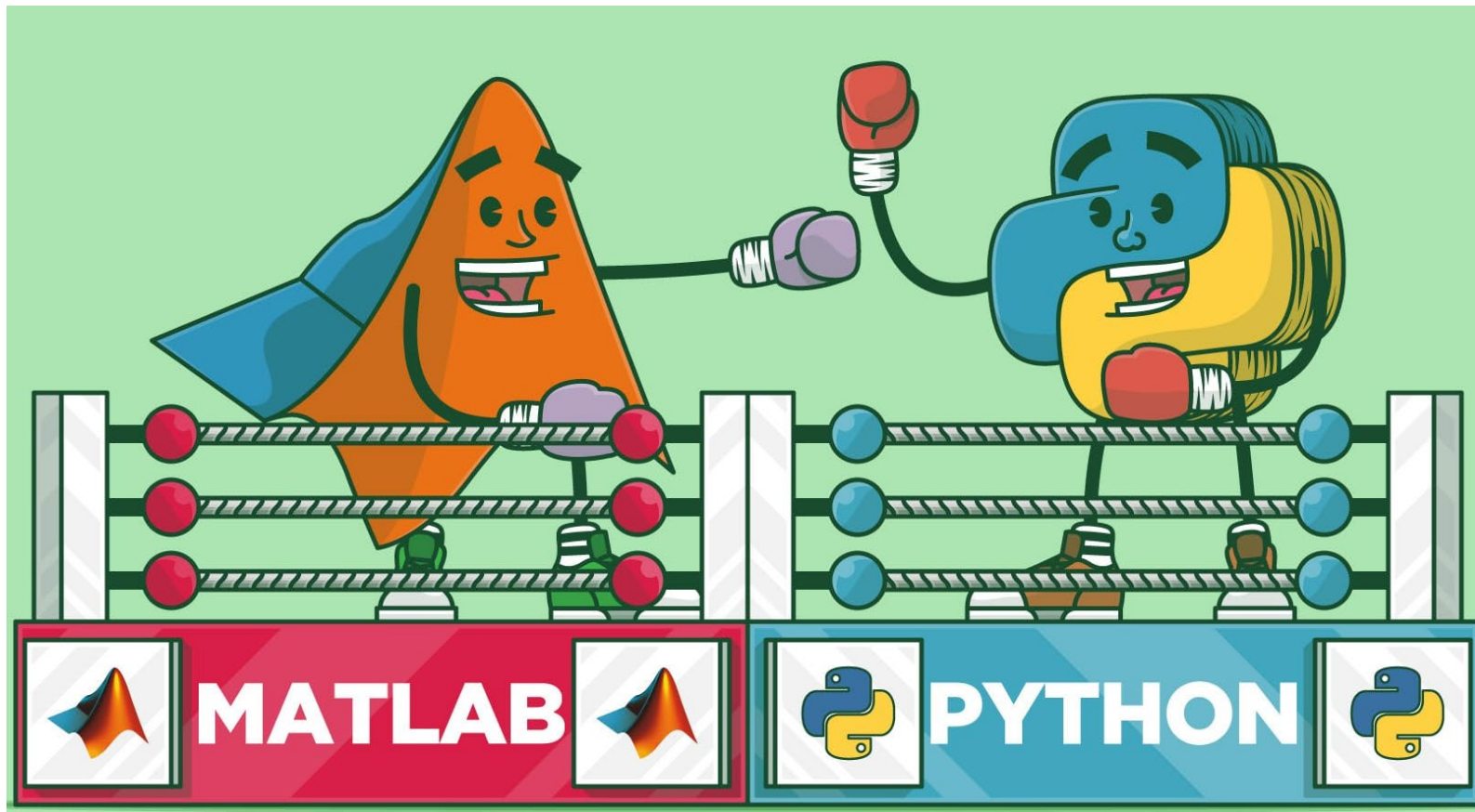
It is possible to verify that $\mathbb{C}^{n \times m}$ with the inner product defined as

$$\langle A, B \rangle := \text{trace}(A^* B) = \sum_{i=1}^n \sum_{j=1}^m \bar{a}_{ij} b_{ij} \quad \forall A, B \in \mathbb{C}^{n \times m}$$

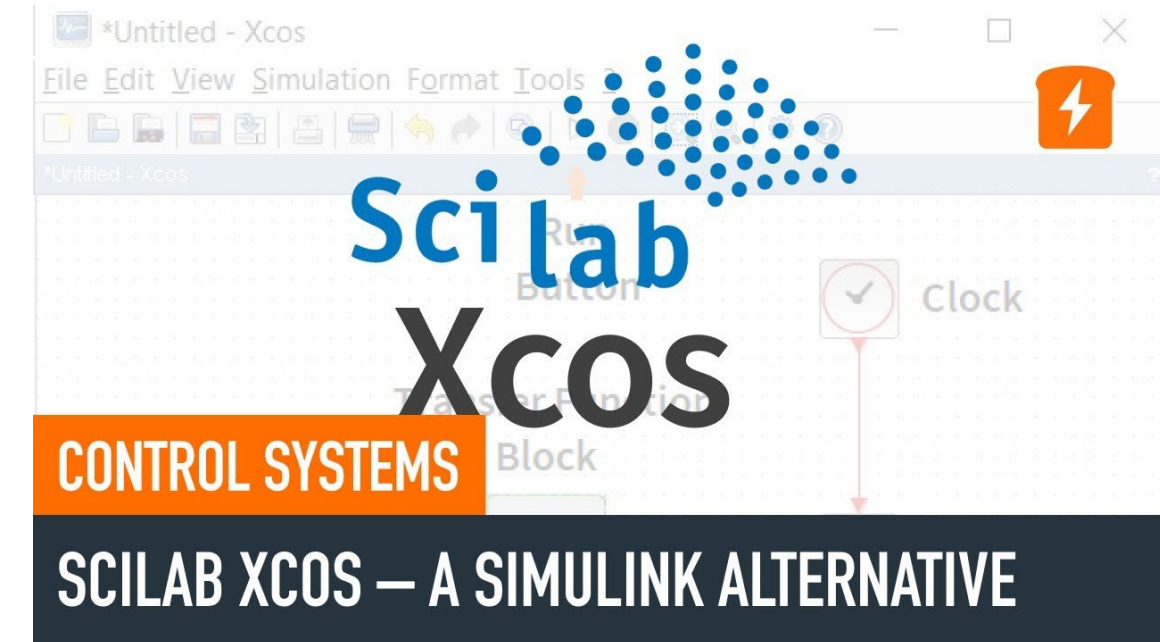
is also a finite-dimensional Hilbert Space.



Computational Tools



julia



[Python](https://python-control.readthedocs.io/en/stable/index.html)

<https://python-control.readthedocs.io/en/stable/index.html>

[Julia](https://juliacontrol.github.io/RobustAndOptimalControl.jl/dev/uncertainty/)

<https://juliacontrol.github.io/RobustAndOptimalControl.jl/dev/uncertainty/>



Matlab Features for RC

Advantages

- It offers specialized, consistent, and reliable tools for robust control.
 - It allows direct and easy handling of systems in transfer functions, state space, SISO or MIMO.
 - It allows for the easy implementation of mathematically complex processes.
- Furthermore, the Matlab toolbox offers up-to-date and convenient documentation.

Disadvantages

- It is a licensed software.

